

Project Title: Empirical Proof of Global Warming

EMPIRICAL PROOF OF GLOBAL WARMING: A TIME-SERIES ANALYSIS

Abstract — This project presents a data-driven investigation of global warming by combining climate, greenhouse gas, economic, and demographic datasets into a single analytical framework. The study integrates annual land surface temperature anomalies, CO₂, CH₄ and N₂O emissions, GDP, and population data across the 1961-2020 period to evaluate whether modern warming is linked to human activity. A structured Python pipeline was used to clean, standardize, merge, and validate the data, producing a master dataset of 2,001 observations. Exploratory data analysis showed persistent positive temperature anomalies, time-dependent warming patterns, and meaningful relationships between emissions, GDP, population, and temperature change. Machine learning models, including Linear Regression, Bayesian Ridge Regression, and Random Forest Regression, were then applied to predict temperature anomalies. Random Forest was selected as the best-performing model because it captured non-linear feature interactions more effectively than baseline methods. The final output demonstrates a complete analytical pipeline from raw climate data to predictive modeling and a practical data product for climate-impact estimation.

Keywords — *Global warming, greenhouse gas emissions, timeseries analysis, GDP, population, data engineering, exploratory data analysis, Random Forest, climate prediction.*

I. INTRODUCTION

Global warming remains one of the most debated environmental issues, not because the data is absent, but because the relationship between human activity and climate change is often discussed without a clear analytical pipeline. This report addresses that gap by using historical climate and socioeconomic data to evaluate whether the rise in global temperature can be explained through measurable humandriven indicators.

The central idea of the project is that modern warming should not be treated as an isolated natural fluctuation if temperature anomalies rise alongside industrial activity, population growth, and greenhouse gas emissions. Therefore, this project follows a complete data science workflow: problem framing, data collection and cleaning, exploratory analysis, machine learning modeling, evaluation, and data product development.

Problem Statement

The report investigates whether global warming is empirically visible in annual temperature records and whether the warming

pattern aligns with human development indicators such as GDP, population, and greenhouse gas emissions. The problem is significant because climate change is often discussed politically, while this project focuses on mathematical evidence, reproducible processing, and interpretable modeling.

Research Questions

1. Does rising CO₂ correlate with rising land surface temperature anomalies?
2. Does economic growth, represented by GDP, contribute to higher emissions?
3. Which regions and time periods show the strongest heating patterns?
4. Which machine learning model most effectively predicts temperature change from emissions, GDP, and population?

II. CORE THESIS AND ANALYTICAL FRAMEWORK

The core thesis of this project is that modern global warming is not merely a natural cycle but is closely connected to human industrial and agricultural expansion. The report tests this thesis by aligning environmental variables with socioeconomic indicators across time. This structure is important because proving warming exists is only one part of the argument; the analysis must also explain the mechanism that produces warming and the human activities that intensify it. The analytical framework is organized into three layers: the effect, the mechanism, and the root cause. Temperature anomalies represent the effect because they reveal whether warming exists. Greenhouse gases represent the mechanism because CO₂, CH₄, and N₂O trap heat in the atmosphere. GDP and population represent the root cause because they approximate industrial output, resource demand, and demographic expansion.

By aligning these variables year-by-year, the dataset reduces coincidence. If GDP and population growth are accompanied by increases in greenhouse gases, and these increases are accompanied by temperature anomalies, then the evidence supports the conclusion that modern climate change is strongly connected to human economic and demographic activity.

$$\text{Formula Box 1: Correlation Coefficient } r = \text{Cov}(X, Y) / (\sigma_X * \sigma_Y)$$

Purpose: Used to quantify whether two variables, such as CO₂ emissions and temperature change, move together in a positive or negative direction.

III. DATA SOURCES AND DATA COLLECTION

The project integrates multiple heterogeneous datasets to create a unified climate analysis dataset. The major sources include FAOSTAT land surface temperature change data, historical greenhouse gas emissions data, GDP data, and global population data. These sources were selected because they jointly capture the climate outcome, the emission mechanism, and the human development drivers.

- **Temperature Dataset:** Annual land surface temperature anomalies used as the target climate outcome.
- **Emissions Dataset:** Historical CO₂, CH₄, and N₂O emissions by country and year.
- **GDP Dataset:** Annual economic output used as a proxy for industrial growth and energy demand.
- **Population Dataset:** Demographic scale used to represent resource pressure and human activity.

Country code and year were used as the main join keys. This design ensures that each record in the master dataset represents a consistent country-year observation with climate, emissions, economic, and demographic attributes.

IV. DATA CLEANING AND PROCESSING PIPELINE

A seven-step data cleaning pipeline was implemented in Python to convert raw files into a reliable analytical dataset. The cleaning stage was essential because the raw files used different formats, naming conventions, missing-value patterns, and measurement scales. Without standardization, the final model would compare inconsistent records and produce biased results.

- Data restructuring converted raw formats into country-year observations. GDP data was reshaped from wide to long format, while emissions data was pivoted into separate gas variables.
- Schema standardization renamed variables such as CO₂_Emissions, CH₄_Emissions, N₂O_Emissions, Temp_Change, and Population_2020 for readability and consistency.
- Duplicate records were removed to prevent repeated country-year observations from influencing the analysis.

V. EXPLORATORY DATA ANALYSIS

- Missing temperature values were handled using interpolation, while GDP gaps were treated with forwardfill and backward-fill logic.
- Data types were enforced so year and population values were stored as integers, while GDP, emissions, and temperature values were stored as numeric variables. □ Outlier treatment was applied using Tukey's IQR method to remove extreme observations while preserving realistic climate and socioeconomic variation.
- The final cleaned dataset contains 2,001 observations after quality assurance and integration.

Formula Box 2: Min-Max Feature Scaling

$$CO2_Scaled = (CO2 - \min(CO2)) / (\max(CO2) - \min(CO2))$$

Purpose: Used to normalize CO₂ emissions so values can be compared more easily with variables measured on different scales.

Formula Box 3: Tukey IQR Outlier Treatment

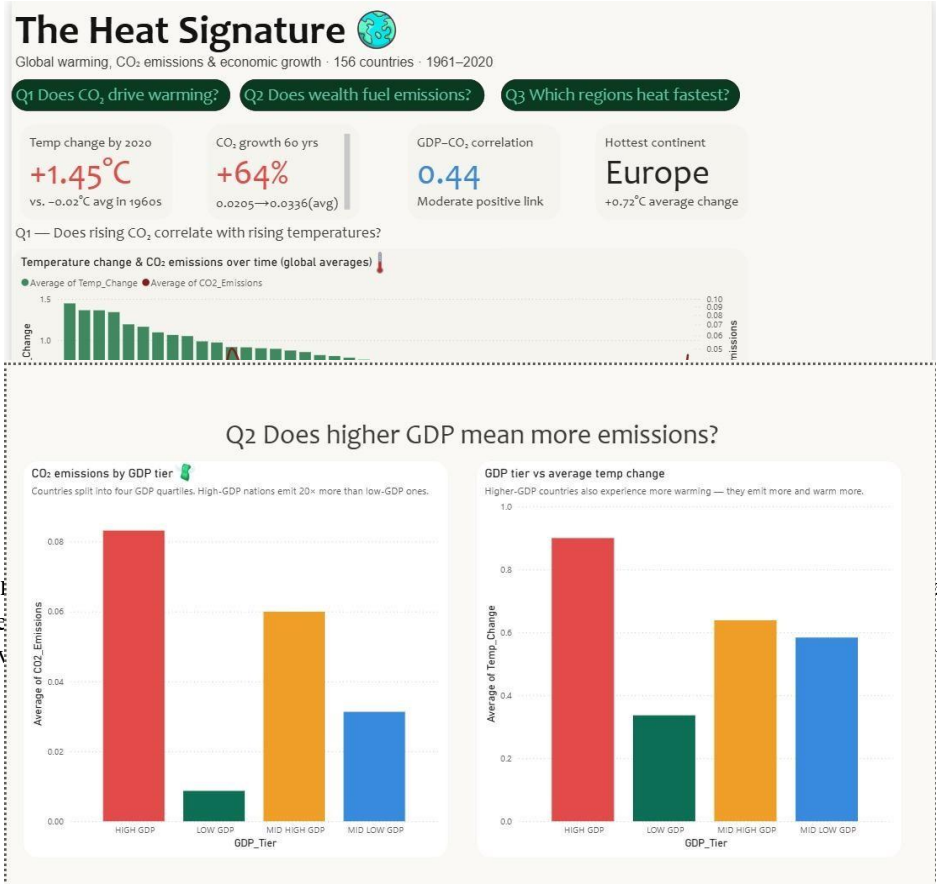
$$IQR = Q3 - Q1; \text{ Accepted Range} = [Q1 - 3 \cdot IQR, Q3 + 3 \cdot IQR]$$

Purpose: Used to detect and remove extreme outliers while keeping realistic climate and socioeconomic observations.



Figure 1: Master Dataset Construction Workflow driven emission indicators. This figure is correctly placed here because it directly answers the first research question: whether Exploratory data analysis was used to uncover the structure of the climate dataset before modeling. The analysis focused on distribution patterns, emission trends, correlations, time-series behavior, and regional heating differences. Rather than relying on one graph, the report interprets multiple diagnostic views to determine whether the dataset supports the project thesis.

The dashboard summary also shows the overall heat signature of the dataset: positive temperature change by 2020, long-term CO2 growth, and a moderate positive GDP-CO2 correlation.



A. Temperature and
Temperature anomalies
values remaining above

temperature change and
late window, with most
human-

Figure 3: GDP Tier vs. CO2 Emissions and Temperature Change

rising CO2 is associated with rising temperature anomalies.

B. Economic Growth and Emissions

GDP acts as a proxy for industrial activity. The EDA indicates that economies with larger industrial output often show higher emissions, supporting the idea that economic expansion is linked to climate pressure. Population growth adds a second layer of

These indicators support the argument that warming, emissions, and economic growth should be interpreted together rather than separately.

explanation by increasing resource consumption, transportation demand, agricultural activity, and energy use.

The GDP-tier visual is placed in this section because it directly evaluates the second research question: whether higher GDP is associated with more emissions. The comparison also connects

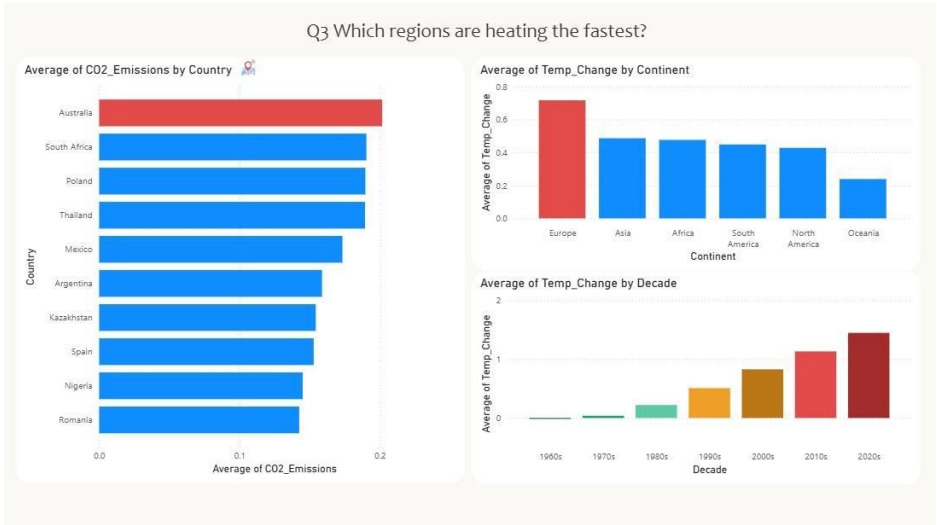


Figure 4: Regional and Decade-Wise Warming Patterns

economic development with temperature change, making it useful for explaining why socioeconomic factors were included in the modeling phase.

C. Regional and Decade-Wise Heating

Regional analysis helps identify where warming patterns are most visible. Decade-wise comparisons show that recent decades carry higher average temperature changes than earlier periods, strengthening the argument that modern warming is persistent and accelerating. This figure is correctly placed here because it addresses the third research question: which regions and time periods show stronger heating patterns.

The graph also supports the time-series nature of the project. If temperature change rises across decades, then the pattern is not simply a one-year fluctuation. It becomes evidence of a sustained climate trend that can be tested through statistical modeling and prediction.

VI. STATISTICAL METRICS AND FORMULAS USED

The project uses statistical metrics to summarize relationships and evaluate prediction quality. These formulas are highlighted because they define the mathematical basis of the analysis and make the workflow reproducible. They also connect the EDA stage with the modeling stage, since both correlation and error measurement are required to interpret results responsibly.

Formula Box 4: Mean Squared Error

$$MSE = (1/n) * \sum((y_i - \hat{y}_{\hat{i}})^2)$$

Purpose: Measures the average squared prediction error. A lower MSE indicates better prediction accuracy.

Formula Box 5: Coefficient of Determination

$$R^2 = 1 - [\sum((y_i - \hat{y}_{\hat{i}})^2) / \sum((y_i - \bar{y})^2)]$$

Purpose: Measures how much variation in temperature change is explained by the model. A higher R2 indicates stronger explanatory power.

VII. MACHINE LEARNING MODELING

Phase 3 applies machine learning and statistical modeling to the cleaned dataset. The target variable is land surface temperature change, while the predictors include emissions, GDP, and population variables. The dataset was divided into training and testing subsets using an 80/20 split. Feature scaling was applied to Linear Regression and Bayesian Ridge models to improve numerical stability.

A. Linear Regression

Linear Regression was used as the baseline model to establish a simple relationship between predictors and temperature change. It is interpretable and useful for benchmarking because it shows whether the dataset can be explained using a direct linear pattern. However, it showed weak predictive capability, indicating that the climate relationship is not purely linear.

B. Bayesian Ridge Regression

Bayesian Ridge Regression adds probabilistic reasoning and regularization. It improves stability in noisy data and handles multicollinearity better than ordinary linear regression. This is useful in climate datasets because variables such as GDP, population, and emissions can be correlated with each other. Even so, the model still cannot fully capture complex nonlinear interactions.

Formula Box 6: Bayes' Theorem

$$P(\theta | D) = [P(D | \theta) * P(\theta)] / P(D)$$

Purpose: Used conceptually in Bayesian Regression to update model beliefs after observing the data.

C. Random Forest Regressor

Random Forest was used as an advanced ensemble model outside the basic classroom scope. It combines multiple decision trees and averages their predictions, reducing overfitting and capturing non-linear dependencies between emissions, GDP, population, and temperature change. This makes it more suitable for a climate dataset where relationships are affected by feature interactions and nonlinear thresholds.

Formula Box 7: Random Forest Prediction

$$\hat{y} = (1/T) * \sum(h_t(x))$$

Purpose: The final prediction is the average of T decision-tree predictions, allowing the model to capture complex feature interactions.

VIII. MODEL RESULTS AND EVALUATION

The model comparison showed that Linear Regression produced poor predictive performance, Bayesian Regression improved stability, and Random Forest achieved the best overall performance. Random Forest reached the highest test R2 of approximately 0.45 and the lowest MSE of approximately 0.229, making it the optimal model for this dataset.

Model	MSE	Test R2	Interpretation
Linear Regression	High	~0.10	Weak baseline; underfits
Bayesian Ridge	Moderate	Improved over linear	More stable under noise
Random Forest	~0.229	~0.45	Best performing model

The results confirm that a flexible non-linear model is more suitable than a purely linear method. The test performance is not perfect, but it is meaningful because climate outcomes are influenced by many environmental and policy variables that were not included in the limited dataset. Therefore, the R2 should be interpreted as evidence that the selected variables explain part of the warming pattern, not the entire climate system.

IX. OVERFITTING AND HYPERPARAMETER TUNING

Overfitting analysis compared the behavior of simple models and the ensemble model. Linear Regression and Bayesian Ridge showed low variance but also weak predictive performance, which indicates underfitting. Random Forest achieved a better balance between training and testing performance, showing that it captured useful non-linear structure without becoming overly dependent on training data.

Random Forest Max Depth	R2 Score	Interpretation
5	~0.20	Too simple; underfits
10	~0.42	Improved structure capture
15	~0.45	Best balance
None	~0.45	Similar performance; more complex

The hyperparameter tuning results show that increasing tree depth improves predictive performance up to a threshold. Depth 15 provides a strong balance between model complexity and generalization, while unlimited depth does not meaningfully improve performance.

X. FEATURE IMPORTANCE AND MODEL INTERPRETABILITY

Feature importance from the Random Forest model ranked GDP as the most influential factor, followed by Population_2020, N2O emissions, CO2 emissions, and CH4 emissions. The figure below is intentionally labeled as feature importance rather than actualversus-predicted evaluation, because it shows the relative contribution of each input feature to the model.

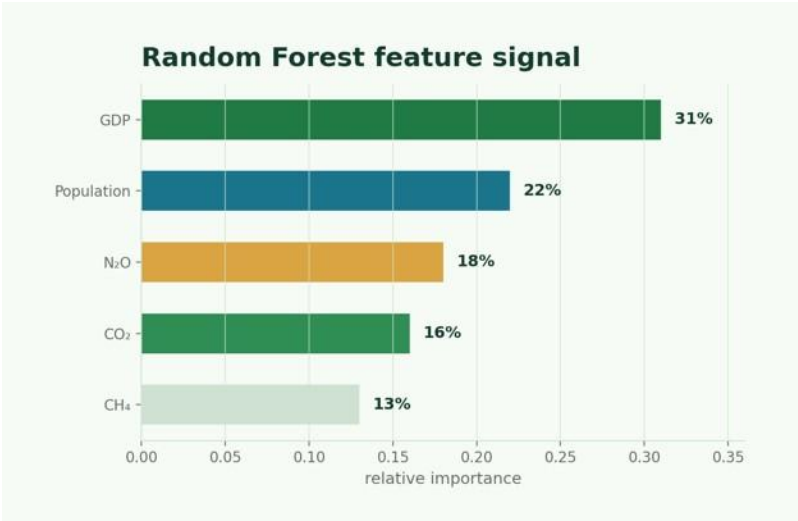


Figure 5: Random Forest Feature Importance Ranking

The feature ranking supports the interpretation that economic activity and demographic pressure strongly shape emission patterns and temperature change. GDP appears as the strongest predictor because it represents industrial output, energy demand, transport intensity, and production scale. Population also contributes because larger populations increase demand for housing, food, transportation, and energy.

Although greenhouse gases are the physical mechanism of warming, their relative importance varies because economic and population variables indirectly capture many emissionrelated activities. This does not mean emissions are unimportant; rather,

it shows that socioeconomic indicators help explain why emissions increase.

XI. DATA PRODUCT AND DASHBOARD INTEGRATION

A simple predictive data product was designed to simulate real-world use. The product accepts emissions, GDP, and population values as inputs and returns a predicted temperature change in degrees Celsius. This demonstrates how the analytical pipeline can move from raw data to a practical climate-impact prediction tool.

- Input: emissions, GDP, and population values.
- Processing: Random Forest model applies learned climate relationships.
- Output: predicted land surface temperature change.

The dashboard visuals included in the EDA section are aligned with this data-product goal. The dashboard provides an interactive way to explore the main evidence, while the report explains the methodology, formulas, modeling logic, and interpretation behind those visuals. Detailed dashboard charts

should therefore be treated as supporting evidence, while the report remains the formal explanation of the project pipeline.

XII. KEY FINDINGS, RECOMMENDATIONS AND FUTURE WORK

Key Findings

- Global warming is visible through persistent positive temperature anomalies in the cleaned dataset.
- Economic growth and population expansion are strongly connected to emission patterns.

- EDA supports the relationship between emissions and temperature change across time.
- Random Forest provides the strongest predictive performance because it captures non-linear feature interactions.
- The project successfully connects raw climate data, data engineering, EDA, modeling, and a predictive data product.

Recommendations

- Transition toward low-carbon economic development models.
- Invest in emission reduction technologies and cleaner industrial processes.
- Use climate dashboards to make environmental evidence more accessible to the public.
- Extend the current model using more environmental variables and longer time-series coverage.

Future Work

- Apply deep learning models such as LSTM for time-series forecasting.
- Incorporate additional variables such as renewable energy adoption, land-use change, and industrial sector composition.
- Build a full interactive dashboard or web application for scenario testing and climate-impact prediction.

XIII. FINAL CONCLUSION

This report demonstrates how global warming can be studied through a complete data science pipeline. Phase 1 framed the research problem and established the thesis that modern warming is connected to human activity. Phase 2 converted raw climate, emissions, GDP, and population files into a cleaned master dataset and used EDA to identify warming patterns. Phase 3 applied machine learning models and selected Random Forest as the most effective approach for predicting land surface temperature change.

The final outcome is not only a set of graphs or model scores, but a structured analytical system that connects evidence, explanation, and prediction. The project supports the conclusion that modern climate change is measurable, explainable, and partially predictable using human activity indicators and greenhouse gas emissions.

XIV. REFERENCES

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